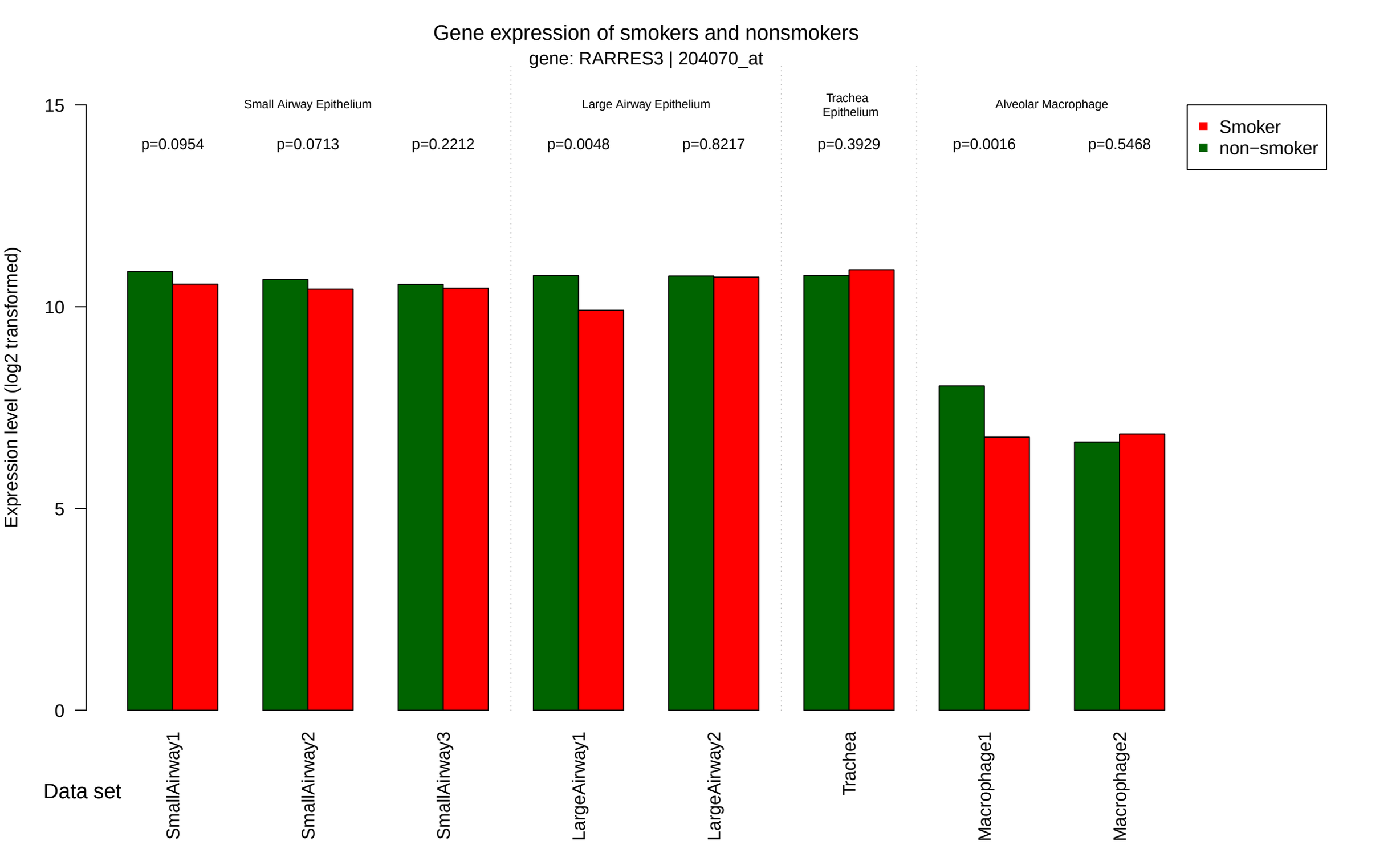
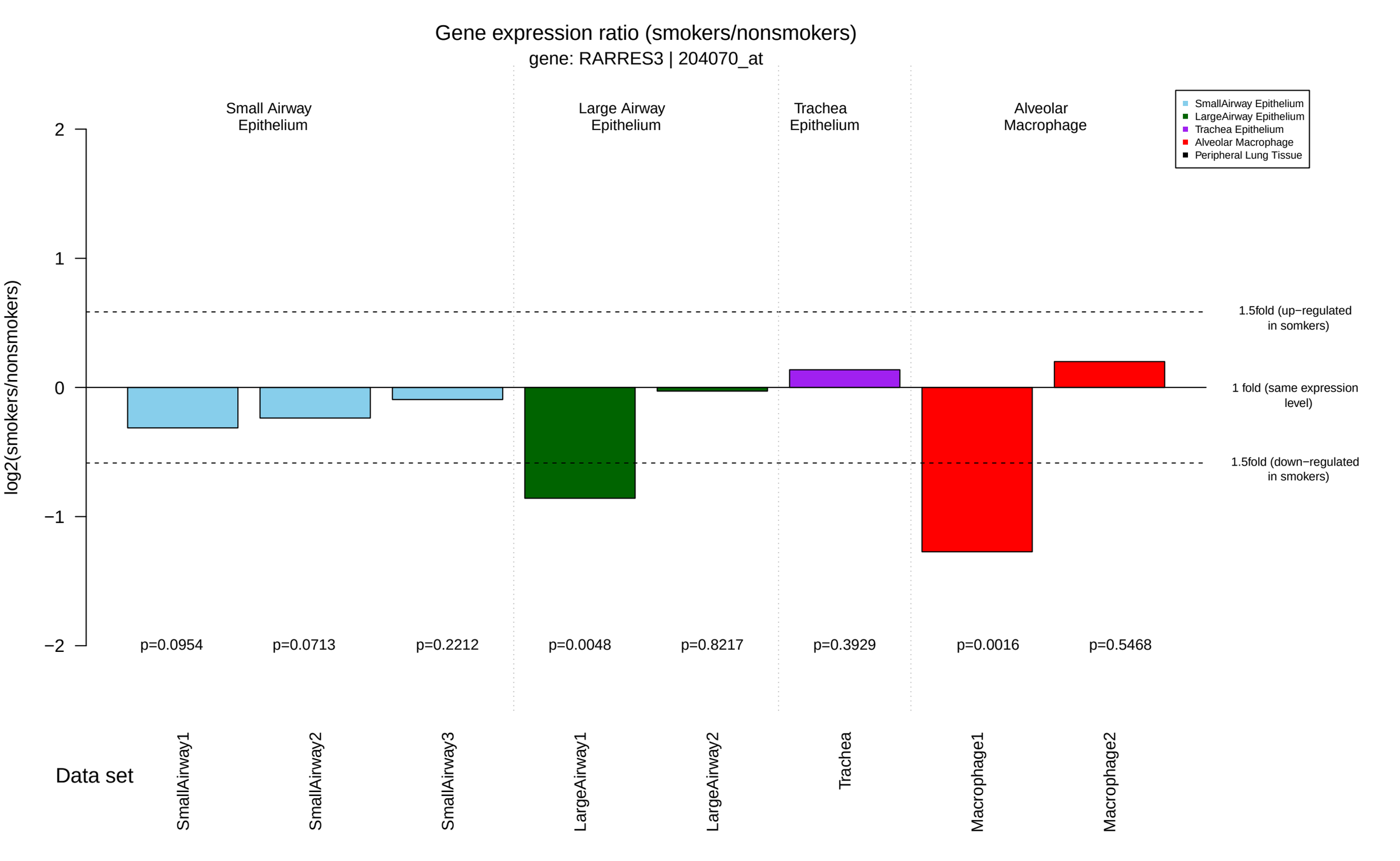
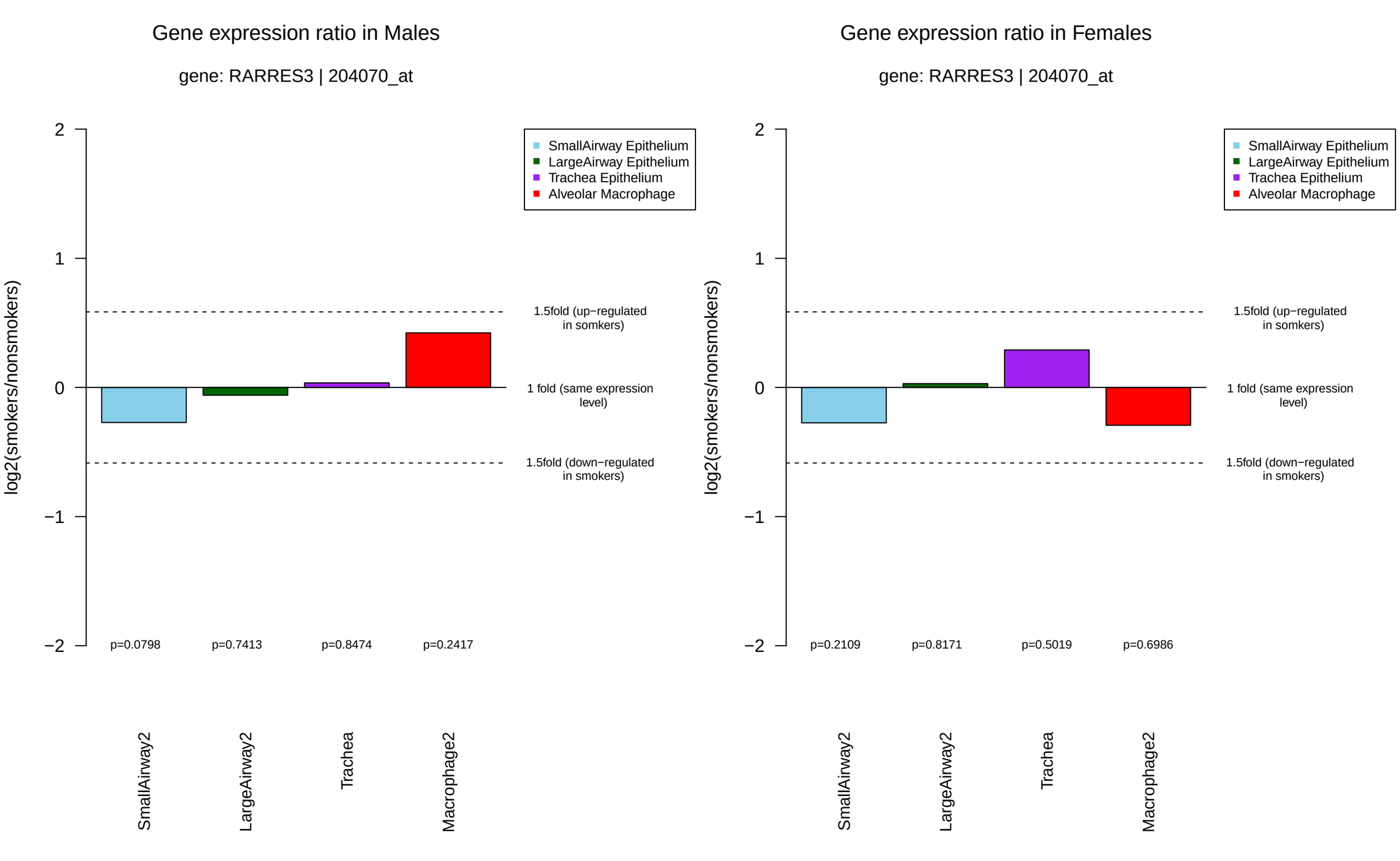
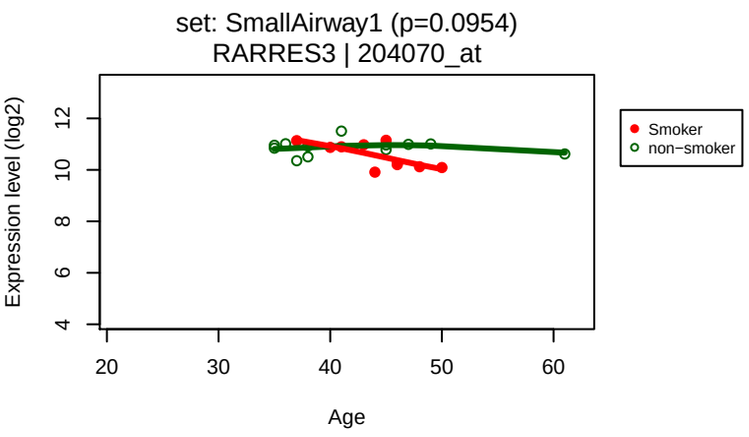
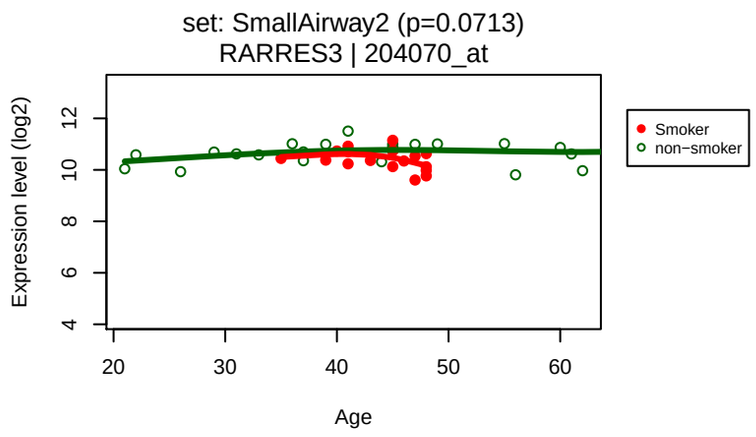
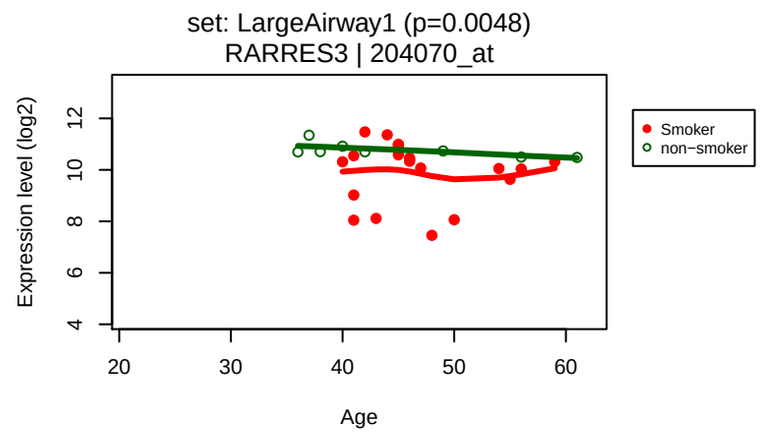

Gene expression of smokers and nonsmokers
gene: RARRES3 | 204070_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0954
p=0.0713
p=0.2212
p=0.0048
p=0.8217
p=0.3929
p=0.0016
p=0.5468
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

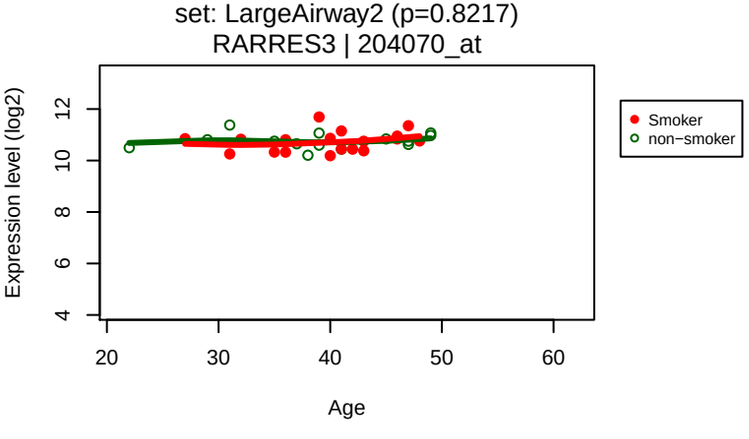

Gene expression ratio (smokers/nonsmokers)
gene: RARRES3 | 204070_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0954
p=0.0713
p=0.2212
p=0.0048
p=0.8217
p=0.3929
p=0.0016
p=0.5468
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

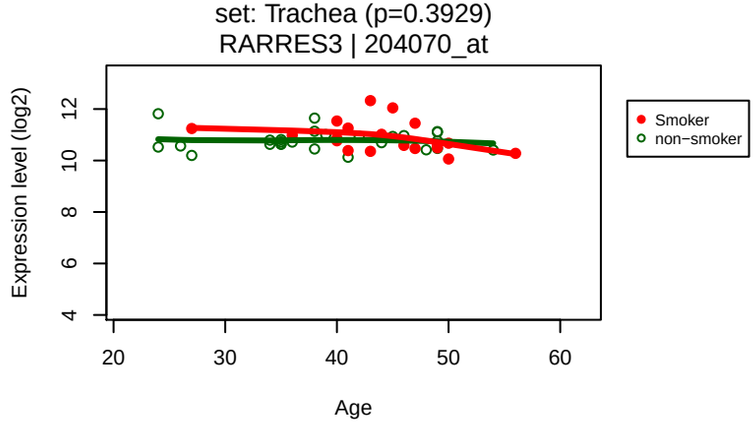

Gene expression ratio in Males
gene: RARRES3 | 204070_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0798
p=0.7413
p=0.8474
p=0.2417
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RARRES3 | 204070_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2109
p=0.8171
p=0.5019
p=0.6986
SmallAirway2
LargeAirway2
Trachea
Macrophage2

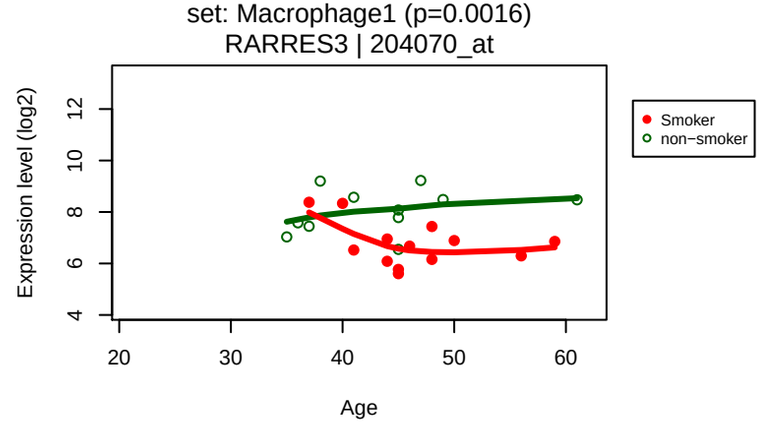

set: SmallAirway1 (p=0.0954)
RARRES3 | 204070_at
Expression level (log2)
Age
Smoker
non−smoker

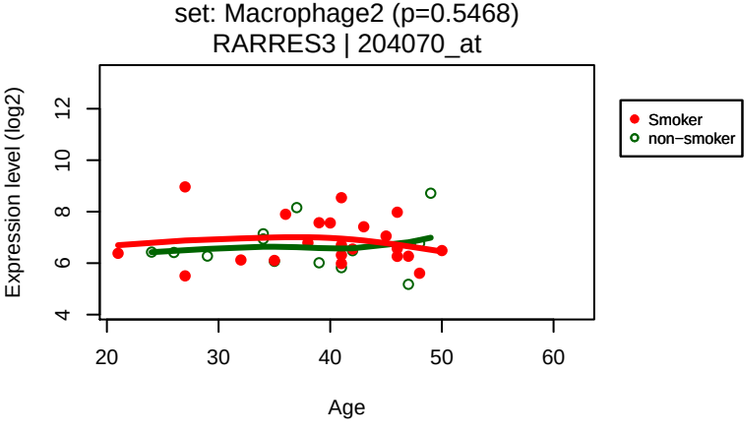

set: SmallAirway2 (p=0.0713)
RARRES3 | 204070_at
Expression level (log2)
Age
Smoker
non−smoker

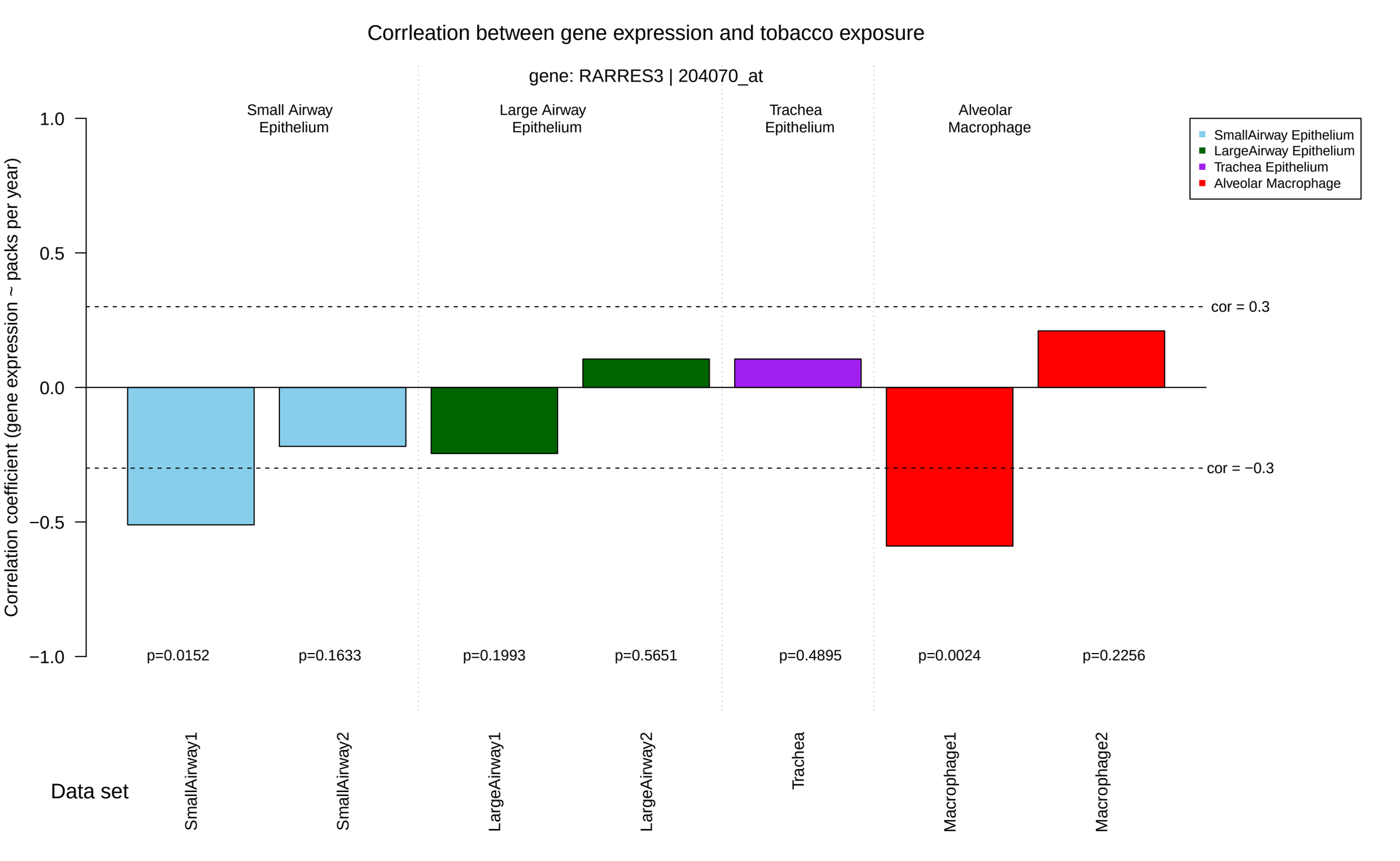

set: LargeAirway1 (p=0.0048)
RARRES3 | 204070_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8217)
RARRES3 | 204070_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3929)
RARRES3 | 204070_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0016)
RARRES3 | 204070_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5468)
RARRES3 | 204070_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RARRES3 | 204070_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0152
p=0.1633
p=0.1993
p=0.5651
p=0.4895
p=0.0024
p=0.2256
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2